Short-term Photovoltaic Power Generation Prediction using Deep Interval Type-2 Fuzzy Model Based on Extended Kalman Filter

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***Abstract*—A forecasting method using deep interval type-2 fuzzy model (DIT2FM) based on extended Kalman filter (EKF) is proposed to tackle the problem of lower forecasting accuracy in the photovoltaic (PV) power generation prediction which is caused by uncertainties of the characteristics of historical PV power sequences. At first, the DIT2FM is constructed in a way of bottom-up and hierarchical connection of many IT2FISs that are taken as its basic building modules. Then, the EKF algorithm is utilized to train the parameters of DIT2FM recursively for the purpose of reducing computational complexity and improving convergence speed. Besides, the proposed method is applied to the short-term prediction of a PV plant's power generation in an area, and compared with other forecasting methods based on deep fuzzy models and deep neural networks under the same conditions. The results of experiments indicate that the proposed DIT2FM method can further improve the prediction accuracy while meeting the actual necessity of real-time dispatching of high percentage of PV-connected power systems by forecasting the PV power generation in an adaptive and online scheme, which brings potential applications.**

Keywords—prediction, PV power generation, deep fuzzy model, interval type-2 fuzzy inference system, extended Kalman filter

I. Introduction

As one of the most commonly and promisingly renewable energy sources, solar energy has been widely used to meet the huge energy demand for all walks of life in the last several decades [1]. Being easily converted into electricity and freely available on the open-air space, such as the rooftop and desert areas much more, solar PV contributes to the highest growth percentage at an average annual rate of 36.0% among global renewable energy sources utilization since 1990 according to the overview of Renewables Information of IEA 2021 [2]. However, the quality of electricity generated by PV systems is highly sensitive to climate and seasonal factors which are extremely stochastic and sophisticated. As a result, the higher PV power penetration will pose enormous challenges for the safety and stability of the grid. To reduce the PV power volatility's impact on the grid, an accurate and reliable PV power output forecasting is acutely required [3].

At present, there have been various models and methods successfully applied for the prediction of short-term PV power generation, which can be largely categorized into physical, statistical and hybrid modeling techniques [4]. Instead of handling the complex coupling mechanisms between different meteorological factors, the statistical technique is compelling in PV power forecasting due to focusing on the actual operation data of the PV plant [5]. Nevertheless, the traditional statistical models typically containing autoregressive integrated moving average (ARIMA), grey box (GB) model, and so forth tend to fail to achieve a satisfying PV power generation prediction accuracy because of their lower ability to deal with non-linear and complex PV datasets. It is for this very limitation that more advanced statistical models namely machine learning (ML) approaches are developed to improve the performance of models in the prediction of PV power generation.

In the wide group of ML, two major representative branches are artificial neural networks (ANN) and fuzzy inference system (FIS) respectively. For example, Kumar et al. used the ANNs to predict the real-time output power of a PV system. To improve the accuracy of PV power generation prediction, a combined technique of ANN and FIS namely adaptive networked-based fuzzy inference system (ANFIS) was applied in. In addition, the support vector machine (SVM) and extreme learning machine (ELM) are also receiving much attention for the PV power generation prediction. With the flourish of artificial intelligence techniques, the forecasting models improved with optimization algorithms (OAs), deep learning methods, and hybrid-based prediction models play their respective roles on the way of fulfilling the higher forecasting accuracy of PV power generation successively.

However, there are more or less drawbacks for the aforementioned forecasting methods, like poor interpretability and largely computational expense of deep neural networks (DNNs), the rule explosion problem of fuzzy inference systems (FISs) when applied to the high-dimensional problems with big data. To handle the above problem, a growing number of research individuals and communities are drawing inspiration from the combination of advantages flowing from the deep learning methods and FISs. For instance, a deep convolutional fuzzy system (DCFS) was constructed through a multilayer connection of many FISs and successfully used in the prediction of the real Hang Seng Index of the Hong Kong stock market. Motivated by the DNN and DCFS.

To summarize the above analysis, it can be concluded that the multilevel fuzzy rule-based DFM have a superiority in the model interpretability while avoiding the fuzzy rule explosion problem. To develop a powerful DFM, the most important thing is to design its training algorithm, just like the Wang-Mendel (WM) fast learning algorithm for training DCFS and the constrained least square method involved in. A good training approach for the antecedent and consequent

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parameters of a FIS is required to reduce the computational complexity and improve convergence speed to the optimal result. Inspired by this, in this paper, an EKF-based training strategy for an IT2FIS is introduced. On the one hand, compared with the existing training algorithms for DFMs, the EKF-based parameters learning method demonstrates its strength in the face of noise and uncertainties in the data, which can provide an optimal parameters estimation in the learning process of a forecasting model. On the other hand, an IT2FIS also has the ability to deal with a higher level of uncertainties than the type-1 FISs (T1FSs) adopted in. On this basis, this study proposes a novel extended Kalman filter based deep interval type-2 fuzzy model (DIT2FM) and applies it to the PV power generation forecasting.

The rest of this paper is organized as follows: In section II, the DIT2FM will be proposed, where its general structure and training algorithm will be described in details. In section III, the comparative experiments will be conducted through a real-world PV power generation dataset; In section IV, the conclusions of this study will be given.

II. Deep Interval Type-2 Fuzzy Model

A. The general structure of DIT2FM

The general structure of a DIT2FM is illustrated in Fig. 1 , where the ***Conv*** as a sliding window (also called convolutional operator) selects the input variables to the fuzzy inference systems (FISs). As depicted in the figure, the proposed DIT2FM in this paper is constructed through an ensemble of many IT2FISs in a bottom-up and layer-by-layer scheme, where the input vector $\boldsymbol{X}^0=(x_1^0,x_2^0,\ldots,x_n^0)$ to the DIT2FM is generally of very high dimension and processed chunk-by-chunk within the IT2FISs in the **Layer** 1 whose integrated output vector $\boldsymbol{X}^1=(x_1^1,x_2^1,\ldots,x_{n^1}^1)$ is taken as the input vector and fed to the **Layer** 2 accordingly, and then the **Layer** l $(l=1,2,\ldots,L)$ is followed by the same procedure until the final scalar output in **Layer** L is generated.

The functional relationship between inputs and outputs of the of DIT2FM is formulated as follows:

$$\boldsymbol{X}^l=F^l(\boldsymbol{X}^{l-1})=(f_1^l(\boldsymbol{x}_1^{l-1}),\cdots,f_{n^l}^l(\boldsymbol{x}_{n^{l-1}}^{l-1})) \tag{1}$$

where $\boldsymbol{X}^{l-1}$ and $\boldsymbol{X}^l$ are the respective input and output vectors of the **Layer** l ; n^l is the number of IT2FISs implemented in **Layer** l ; $\boldsymbol{x}_1^{l-1},\cdots,\boldsymbol{x}_{n^l}^{l-1}$ are the n^l sub-features split from $\boldsymbol{X}^{l-1}$ by sliding windows $\boldsymbol{Conv}_1^l,\cdots,\boldsymbol{Conv}_{n^l}^l$, $f_{n^l}^l(\bullet)$ and $F^l(\bullet)$ represents the n^lth IT2FIS and aggregate function in **Layer** l , respectively.

Consider that the output of the **Layer** l serves as the input of the **Layer** $l+1$ (namely, $\boldsymbol{X}^{l+1}\leftarrow\boldsymbol{X}^l$), the DIT2FM can be mathematically modeled by the following composite function:

$$x^L=(F^L\circ F^{L-1}\circ\cdots\circ F^1)(\boldsymbol{X}^0) \tag{2}$$

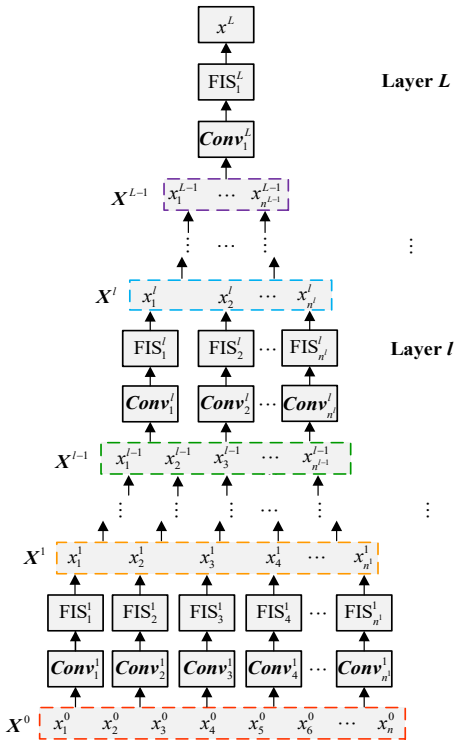

Fig. 1 General architecture of DIT2FM.

Before applying the DIT2FM, it is a must to determine the parameter settings of the ensemble architecture of the DIT2FM with regard to number of layers L and arguments of the sliding window ***Conv*** used for each layer including window size and step size according to the dimension of the input vector to each layer, that means the number of IT2FISs of each layer is adaptive and changeable.

The structure of an IT2FIS is illustrated in the Fig. 2, which is a five-layer, multi-input-single-output (MISO) network structure. The fuzzy rule antecedent and consequent parts of a IT2FIS are compiled of IT2 fuzzy sets and crisp numbers respectively, where the consequent parts are TSK type, abbreviated as TSK/A2-C0, and the jth rule is expressed as follows

$$\begin{aligned}R_j:\ &\text{IF } x_1 \textit{ is } \tilde{A}_{j1} \text{ and } x_2 \text{ is } \tilde{A}_{j2} \text{ and } \ldots \text{and } x_n \text{ is } \tilde{A}_{jn},\\ &\text{THEN } u_j=\sum_{i=1}^{n} w_{ij}x_i+b_j\end{aligned} \tag{3}$$

where $x_i(i=1,\cdots,n)$ are the input variable, $u_j(j=1,\cdots,M)$ are the output variable; $\tilde{A}_{ji}$ represents interval type-2 fuzzy sets for the jth rule of the ith input, the corresponding membership function is expressed as (4); w_{ij} and b_j are the parameters in the consequent part of the jth rule.

$$\mu_{\tilde{A}_{ji}}(x)=\begin{cases}\left(1-\left|\dfrac{x_i-c}{d}\right|^a\right)^{\frac{1}{a}} & ,a_2<a<a_1 \quad \text{If } |x_i-c|<d\\ 0 & \text{otherwise}\end{cases} \tag{4}$$

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where the parameter a_1 and a_2 determine the width of the uncertainty of the Ellipsoidal MF, to be selected as $a_1 > 1$ and $0 < a_2 < 1$; c and d are the center and the width of the Ellipsoidal membership function (MF).

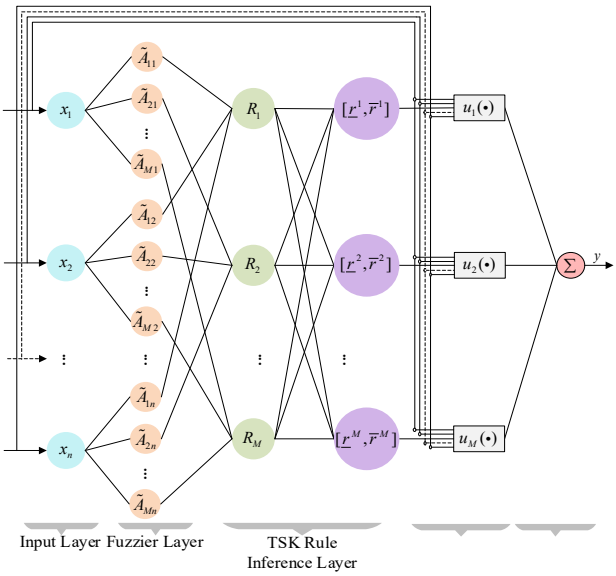

Fig. 2 The structure of TSK/A2-C0 IT2FIS.

In Fig.2, the function $u_j(\bullet)$ includes w_{ij} as the multiplication of the input variables and b_j as the bias values. The final output of TSK/A2-C0 IT2FIS is calculated as

$$y = \int_{f^1 \in [\underline{f}^1, \overline{f}^1]} \cdots \int_{f^1 \in [\underline{f}^M, \overline{f}^M]} 1 \Big/ \frac{\sum_{j=1}^{M} f^j u_j}{\sum_{j=1}^{M} f^j} \tag{5}$$

where $\underline{f}^j$ and $\overline{f}^j$ are given by

$$\begin{aligned} \underline{f}^j(x) &= \underline{\mu}_{\tilde{A}_{j1}}(x_1) * \cdots * \underline{\mu}_{\tilde{A}_{jn}}(x_n) \\ \overline{f}^j(x) &= \overline{\mu}_{\tilde{A}_{j1}}(x_1) * \cdots * \overline{\mu}_{\tilde{A}_{jn}}(x_n) \end{aligned} \tag{6}$$

In which $*$ represents the product t-norm. The output of the fuzzy system in closed form is obtained by

$$\begin{aligned} y &= \frac{\sum_{j=1}^{M} \underline{f}^j u_j}{\sum_{j=1}^{M} \underline{f}^j + \sum_{j=1}^{M} \overline{f}^j} + \frac{\sum_{j=1}^{M} \overline{f}^j u_j}{\sum_{j=1}^{M} \underline{f}^j + \sum_{j=1}^{M} \overline{f}^j} \\ &= \frac{\sum_{j=1}^{M} (\underline{f}^j + \overline{f}^j) u_j}{\sum_{j=1}^{M} \underline{f}^j + \sum_{j=1}^{M} \overline{f}^j} \end{aligned} \tag{7}$$

Thus, the fired strength of the jth rule is defined as follows

$$r^j = \overline{r}^j + \underline{r}^j \tag{8}$$

where $\underline{r}^j$ and $\overline{r}^j$ are given by

$$\begin{aligned} \underline{r}^j &= \frac{\underline{f}^j}{\sum_{j=1}^{M} \underline{f}^j + \sum_{j=1}^{M} \overline{f}^j} \\ \overline{r}^j &= \frac{\overline{f}^j}{\sum_{j=1}^{M} \underline{f}^j + \sum_{j=1}^{M} \overline{f}^j} \end{aligned} \tag{9}$$

B. The training algorithm of DIT2FM

This subsection will present the training algorithm for the DIT2FM. Because every IT2FIS in the DIT2FM follows the exact same learning process, only one single IT2FIS's learning method is described below.

By viewing the training process of an IT2FIS as the parameter estimation problem, the mapping relationship between the output result and input features of the IT2FIS, defined as in (7), can be described by the following nonlinear state-space model

$$\begin{aligned} \boldsymbol{\theta}_{k+1} &= \boldsymbol{\theta}_k + \boldsymbol{\omega}_k \\ y_k &= f_k(\boldsymbol{\theta}_k, \boldsymbol{x}_k) + \upsilon_k \end{aligned} \tag{10}$$

Where $\boldsymbol{\theta}_k$ specifies that the state vector of the IT2FIS at the moment k; y_k represents the IT2FIS's desired output; $f_k(\bullet)$ is a nonlinear function of the input vector $\boldsymbol{x}_k$ and the state vector $\boldsymbol{\theta}_k$; $\boldsymbol{\omega}_k$ and υ_k are the two noise sequences which are independent of each other and typically characterized as white noise with covariance given by

$$\begin{aligned} E[\boldsymbol{\omega}_k] &= 0 \\ E[\boldsymbol{\omega}_k \boldsymbol{\omega}_l^T] &= \delta_{kl} \boldsymbol{Q}_k \\ E[\upsilon_k] &= 0 \\ E[\upsilon_k \upsilon_l^T] &= \delta_{kl} R_k \end{aligned} \tag{11}$$

where $E(\bullet)$ is the expectation operator and δ_{kl} is the Kronecker delta which is defined by

$$\delta_{kl} = \begin{cases} 0 & \text{if } k \neq l \\ 1 & \text{if } k = l \end{cases} \tag{12}$$

To apply the EKF to the state-space model, the nonlinear function $f_k(\bullet)$ given in (10) must be linearized. Therefore, the first-order Taylor expansion is performed to linearize the measurement equation around the current estimate of state vector, expressed by

$$f_k(\boldsymbol{\theta}_k, \boldsymbol{x}_k) = f_k(\hat{\boldsymbol{\theta}}_k, \boldsymbol{x}_k) + F_k \cdot (\boldsymbol{\theta}_k - \hat{\boldsymbol{\theta}}_k) + H.O.T \tag{13}$$

$$\boldsymbol{F}_k^T = \left. \frac{\partial f_k(\boldsymbol{\theta}_k, \boldsymbol{X}_k)}{\partial \boldsymbol{\theta}_k} \right|_{\boldsymbol{\theta}_k = \hat{\boldsymbol{\theta}}_k} \tag{14}$$

where the Jacobian matrix $\boldsymbol{F}_k$ is evaluated at $\boldsymbol{\theta}_k = \hat{\boldsymbol{\theta}}_k$ with $\hat{\boldsymbol{\theta}}_k$ being the estimate of the state vector $\boldsymbol{\theta}_k$ and computed by the EKF at time k, given the observed data up to time $k-1$,

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i.e. y_{k-1}.

Neglecting the higher-order terms ($H.O.T$) the system in (13), the state space model in (10) can be rewritten as

$$\begin{aligned} \boldsymbol{\theta}_{k+1} &= \boldsymbol{\theta}_k + \boldsymbol{\omega}_k \\ y_k &= \boldsymbol{F}_k^T \boldsymbol{\theta}_k + \upsilon_k + \varphi_k \end{aligned} \tag{15}$$

Where φ_k is defined by

$$\varphi_k = f_k\left(\hat{\boldsymbol{\theta}}_k, \boldsymbol{x}_k\right) - \boldsymbol{F}_k^T \cdot \hat{\boldsymbol{\theta}}_k \tag{16}$$

Up to this point, the EKF solution to the parameter estimation problem is given by the following as

$$\begin{aligned} \boldsymbol{K}_k &= \boldsymbol{P}_k \boldsymbol{F}_k \left(R_k + \boldsymbol{F}_k^T \boldsymbol{P}_k \boldsymbol{F}_k\right)^{-1} \\ \hat{\boldsymbol{\theta}}_{k+1} &= \hat{\boldsymbol{\theta}}_k + \boldsymbol{K}_k e_k \\ \boldsymbol{P}_{k+1} &= \boldsymbol{P}_k - \boldsymbol{K}_k \boldsymbol{F}_k^T \boldsymbol{P}_k + \boldsymbol{Q}_k \end{aligned} \tag{17}$$

where $\boldsymbol{K}_k$ is the Kalman gain, $\boldsymbol{P}_k$ is the covariance matrix of the state estimation error and the output estimation error e_k is defined as

$$e_k = y_k - f_k\left(\hat{\boldsymbol{\theta}}_k, \boldsymbol{x}_k\right) \tag{18}$$

In the EKF algorithm, $\boldsymbol{Q}_k$ and R_k , are specified for all training samples, and $\boldsymbol{P}_k$ is initialized at the beginning of training as follows

$$\begin{aligned} \boldsymbol{R}_0 &= \eta^{-1} \boldsymbol{I} \\ \boldsymbol{Q}_0 &= q \boldsymbol{I} \\ \boldsymbol{P}_0 &= \varepsilon^{-1} \boldsymbol{I} \end{aligned} \tag{19}$$

where the parameters η , q , and ε are properly chosen as small scalars which are much less than unity.

When applying the EKF algorithm to an IT2FIS, all the unknown parameters of the IT2FIS are gathered in a vector, which will cause a huge computational cost. To avoid this phenomenon, the parameters vector of the rule premise and consequent parts $\boldsymbol{\theta}_k$ is decomposed into five groups. The parameters updating equations for the parameters in the gth group is as shown below

$$\begin{aligned} \boldsymbol{K}_k^g &= \boldsymbol{P}_k^g \boldsymbol{F}_k^g \left(R_k^g + \left(\boldsymbol{F}_k^g\right)^T \boldsymbol{P}_k^g \boldsymbol{F}_k^g\right)^{-1} \\ \hat{\boldsymbol{\theta}}_{k+1}^g &= \hat{\boldsymbol{\theta}}_k^g + \boldsymbol{K}_k^g e_k \\ \boldsymbol{P}_{k+1}^g &= \boldsymbol{P}_k^g - \boldsymbol{K}_k^g \left(\boldsymbol{F}_k^g\right)^T \boldsymbol{P}_k^g + \boldsymbol{Q}_k^g \end{aligned} \tag{20}$$

Where each group of parameters has its own Kalman parameters $\boldsymbol{K}, \boldsymbol{P}, R, \boldsymbol{Q}, \boldsymbol{H}$.

For a IT2FIS with n inputs, m membership functions for each input and M number of rules in total, as shown in Fig.2, the comprised parameters of these five groups are given by, respectively

$$\begin{aligned} \boldsymbol{\theta}_k^1 &= \left[a_{11}, \cdots, a_{1nm}\right]^T \\ \boldsymbol{\theta}_k^2 &= \left[a_{21}, \cdots, a_{2nm}\right]^T \\ \boldsymbol{\theta}_k^3 &= \left[c_1, \cdots, c_{nm}\right]^T \\ \boldsymbol{\theta}_k^4 &= \left[d_1, \cdots, d_{nm}\right]^T \\ \boldsymbol{\theta}_k^5 &= \left[\omega_{11}, \cdots, \omega_{n1}, \cdots, \omega_{1M}, \cdots, \omega_{nM}, b_1, \cdots, b_M\right]^T \end{aligned} \tag{21}$$

Meanwhile, the corresponding Jacobian matrix $\boldsymbol{F}_k^g$ is defined as

$$\boldsymbol{F}_k^g = \frac{\partial f_k}{\partial \theta_k^g} \tag{22}$$

The main learning process of an IT2FIS is outlined in the form of pseudo-code given by TABLE I.

TABLE I. THE LEARNING PROCESS OF AN IT2FIS

Algorithm. IT2FIS learning procedure
Input: Given N input–output pairs:
$\left[x_{1k}, x_{2k}, \ldots, x_{nk}; y_k\right], \quad k = 1, 2, \cdots N$
\\Phase1: parameters setting for an IT2FIS
Initiate the number of rules M , parameters of the IT2MF a_1, a_2, c, d in (4), and parameters of the consequent part for TSK-type rules w_{ij}, b_j for the jth rule ($i = 1, \cdots, n;\ \ j = 1, \cdots, M$) in (3)
Generate the initial parameters vectors $\boldsymbol{\theta}_k^g\,(g = 1, \ldots, 5)$ in (21)
\\Phase 2: Parameter Adjustment using EKF
for iteration $= 1$ to epochs
Initialize the $\boldsymbol{P}_k^g, \boldsymbol{Q}_k^g, R_k^g$ in (20) via (19)
for $k = 1$ to N
Compute the firing strength r^j for all rules using (4), (6), (8), and (9)
Calculate the final output of an IT2FIS via (7)
Get the output estimation error e_k using (18)
Do EKF-based adjustment mechanism via (20)
end for
end for

III. EXPERIMENTS AND COMPARISONS

A. Comparative models and Evaluation Indices

To prove that the proposed DIT2FM has an advantage in forecasting the PV power generation, the same type of deep fuzzy models DCFS, DIRM-DFM, the similarly ensemble model stacked autoencoders (SAEs), deep convolutional neural network (DCNN), long short-term memory (LSTM) and gated recurrent unit (GRU) neural networks will be involved in comparison with the DIT2FM.

Additionally, to compare the performance of different PV power prediction models mentioned above, four evaluation indicators for the prediction accuracy are applied to evaluate model performance, which are root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error(MAPE) and R-squared (R^2), respectively.

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$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2} \tag{23}$$

$$\mathrm{MAE} = \left|\frac{1}{N}\sum_{i=1}^{N}\left|y_i - \hat{y}_i\right|\right| \tag{24}$$

$$\mathrm{MAPE} = \frac{100\%}{N}\sum_{i=1}^{N}\frac{\left|y_i - \hat{y}_i\right|}{y_i} \tag{25}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{N}\left(y_i - \overline{y}_i\right)^2} \tag{26}$$

Where N is the number of training or testing data samples, y_i is the real PV power value, $\hat{y}_i$ is the forecasted PV power value and $\overline{y}_i$ is the mean of the real PV power values. The better accuracy of the prediction model will be achieved if the values of RMSE and MAE are smaller. If there are excessive outliers or the visibly forecasting delay problem in the prediction, the value of MAPE will be more than 1, which means the prediction model may be not suitable to forecast the nonstationary PV power series. For the indicators $\mathrm{R}^2 \in [0,\ 1]$, the closer the value is to 1, the stronger fitting performance of the prediction model will be.

B. Dataset and configuration of the prediction models

In this experiment, the PV power data in Belgium is applied to measure the performance of the forecasting models, which totally contains 17,660 samples ranged from March 1, 2019 to August 31, 2019 with the sampling frequency being 15min (last 10 days for testing). The Monitored solar PV capacity in Belgium is 4,788 MWp which is a sum of 11 PV plants. This dataset can be downloaded from: https://www.elia.be/nl.

In this experiment, a forecasting window of size 5 is selected, i.e. the four time steps of PV series before time t will be fed to models for the prediction of PV power output at time t. In order to fairly evaluate the forecasting performance of each model, the detailed hyper parameters configuration items list in Table II are determined by trial-and-error, following the guideline that the prediction models can achieve a good forecasting accuracy as well as good regularization, and be compared in an acceptable computational cost, in which the specifics of parameters setting are as follows:

1) Layers: the number of layers for forecasting models (hidden layers for SAEs, LSTM, GRU DCNN);

2) Sub-features: the number of features selected by a sliding window in different Layers (no Sub-features for SAEs, LSTM, GRU);

3) Fuzzy sets/Neuro units: The number of fuzzy sets in each single fuzzy system modules or neuro units in hidden layers;

4) Epochs: The number of training iterations.

TABLE II. THE DETAILED HYPER PARAMETERS CONFIGURATION ITEMS FOR ALL PREDICTION MODELS

Model	Layers	Sub-features	Fuzzy sets/ Neuro units	Epochs
DIT2FM	3	2/3/4	2/2/2	--
DCFS	3	2/3/4	20/20/20	--
DIRM-DFM	3	2/3/4	20/20/20	--
SAEs	3	--	100/100/100	1000
LSTM	3	--	64/32/16	1000
GRU	3	--	64/32/16	1000
DCNN	3	2/2/2	64/32/16	1000

Beyond that, the type of fuzzy set is chosen to be triangular-type for all deep fuzzy models, except the proposed DIT2FM. For SAEs, DCNN, LSTM, GRU and DCNN, using the Tensorflow GPU 1.5.0 calculation platform for the implementations, the activation function for each hidden node is uniformly defined as sigmoid, the model training are optimized by the Rmsprop optimizer, the batch size over each epoch is selected as 256, and the MSE metric is used to measure the loss function. To overcome the overfitting issue in deep neural networks, a 20% dropout rate is adopted, which can get a better anti-overfitting performance through the validation of simulations. Because the estimated results of the deep neural networks usually depend on the initialization of internal weights, the average forecasting errors in terms of four evaluation indicators are calculated over 10 simulations.

C. Experimental results

Table III enlists the RMSE, MAE, MAPE and R2 values obtained for all comparative models over testing datasets. It is evident that the forecasting accuracy of DIT2FM is higher than other DFMs and DNNs.

To further visualize forecasting accuracy of all models, Fig.3 gives the predicted curves of all comparative models over four typical weather types which are sunny, mostly cloudy, rainy and partly cloudy day, respectively. It can be seen that that the DIT2FM can better describe the fluctuations of the actual PV power data.

TABLE III. PERFORMANCE COMPARISONS OF THE PREDICTION MODELS

Model	RMSE(MW)	MAE(MW)	MAPE(%)	R^2
DCNN	26.82	16.28	14.07	0.9901
LSTM	21.87	12.54	22.72	0.9934
GRU	21.44	12.35	24.57	0.9937
SAEs	24.15	15.70	17.15	0.9920
DIRM-DFM	23.85	13.89	16.57	0.9923
DCFS	23.65	15.77	15.42	0.9922
DIT2FM	19.07	10.42	8.03	0.9950

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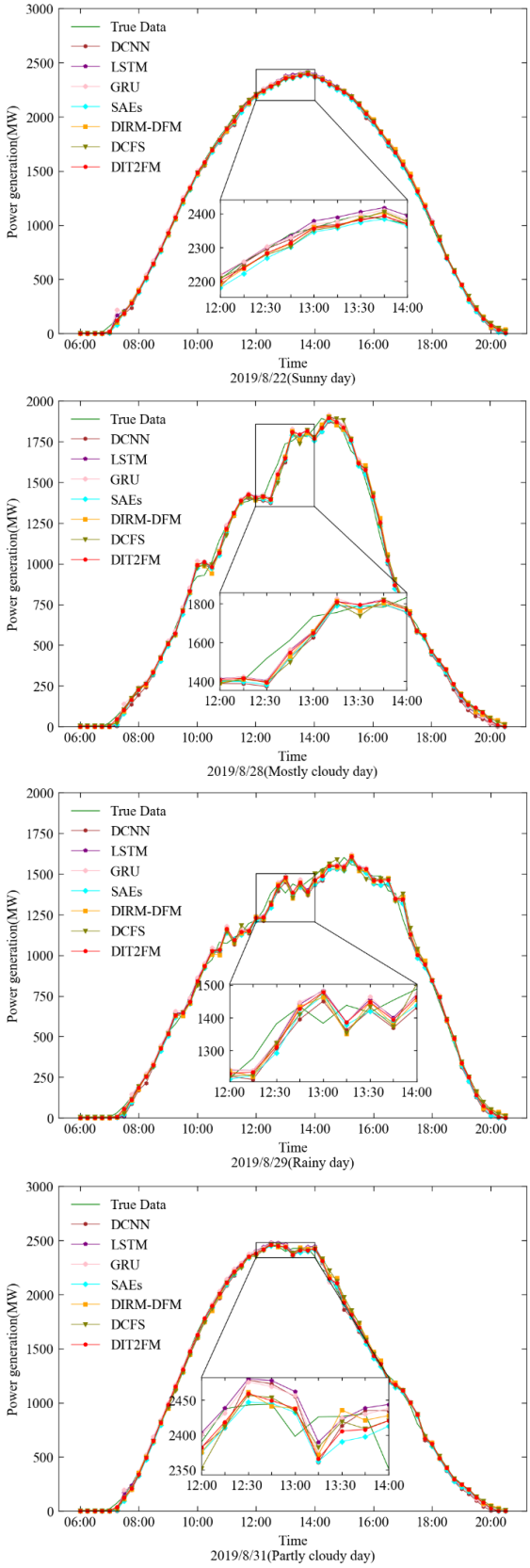

Fig. 3 The predicted curves of all comparative models over four typical weather types.

IV. Conclusion

In this study, a novel DIT2FM is proposed for the prediction of short-term PV power generation. This model is built through the IT2FISs as its building modules by a bottom-up and layer-by-layer scheme. For adopting the ellipsoidal type-2 fuzzy membership function whose biggest uncertainty in the membership value is usually in the mid-range, such IT2FISs can have a better capability to deal with a high level of uncertainties and output more accurate forecasting results. The learning process of a single IT2FIS is built on the EKF-based learning method through which all IT2FISs within the DIT2FM model can work collaboratively to process input features and generate the final output. This assures that the DIT2FM can be trained in an online pattern and excellently approximate to the highly non-linear systems through parallelizable computation.

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